



Faculty of Engineering & Technology

Department of Electrical & Computer Engineering

ENCS5323, Wireless & Mobile Networks

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Course Project

**Real World Spectrum Sensing and Controlled Radio Frequency Analysis Using
Software Defined Radio**

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We used two ADALM-PLUTO units: one in receive mode (`spectrum_sensor.py`) and one in transmit mode (`interference_generator.py`). The scan range is configured in code (`DISP_START_MHZ / DISP_END_MHZ`), e.g. 900–2500 MHz for the full scan and 2400–2500 MHz for Wi-Fi. RX gain was 40–50 dB; interference was band-limited noise at 2412 MHz for 45 s per run. Both programs can run independently (sense only or transmit only) or together (TX on Laptop 2 while RX displays live spectrum on Laptop 1).



Component 1 – Full Spectrum Scan

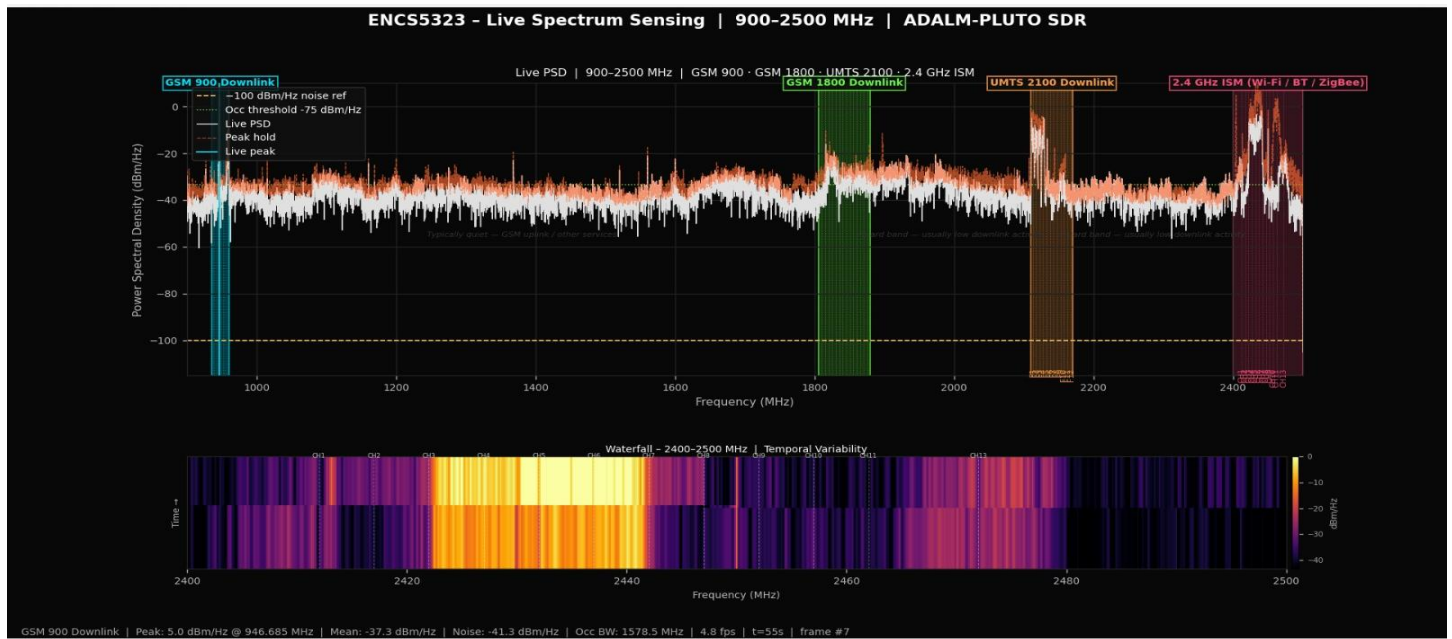


figure 1: Full spectrum PSD (900–2500 MHz) showing GSM 900, GSM 1800, UMTS 2100, and 2.4 GHz ISM activity.

We measured the full downlink range from 900 to 2500 MHz using the Pluto SDR. Figure 1 shows four main active regions: GSM 900 (~935–960 MHz), GSM 1800 (~1805–1880 MHz), UMTS 2100 (~2110–2170 MHz), and the 2.4 GHz ISM band (2400–2500 MHz). The strongest peak in this capture was near 947 MHz (GSM 900). UMTS 2100 and the ISM band also showed clear activity. The gaps between these bands were much quieter, with a noise floor around -40 dBm/Hz. The spectrum changes over time; the waterfall shows bursty Wi-Fi traffic on Channels 6 and 11, while cellular bands (GSM/UMTS) also vary in the PSD above.

Component 1 - Baseline Results (2400–2500 MHz, no transmitter)

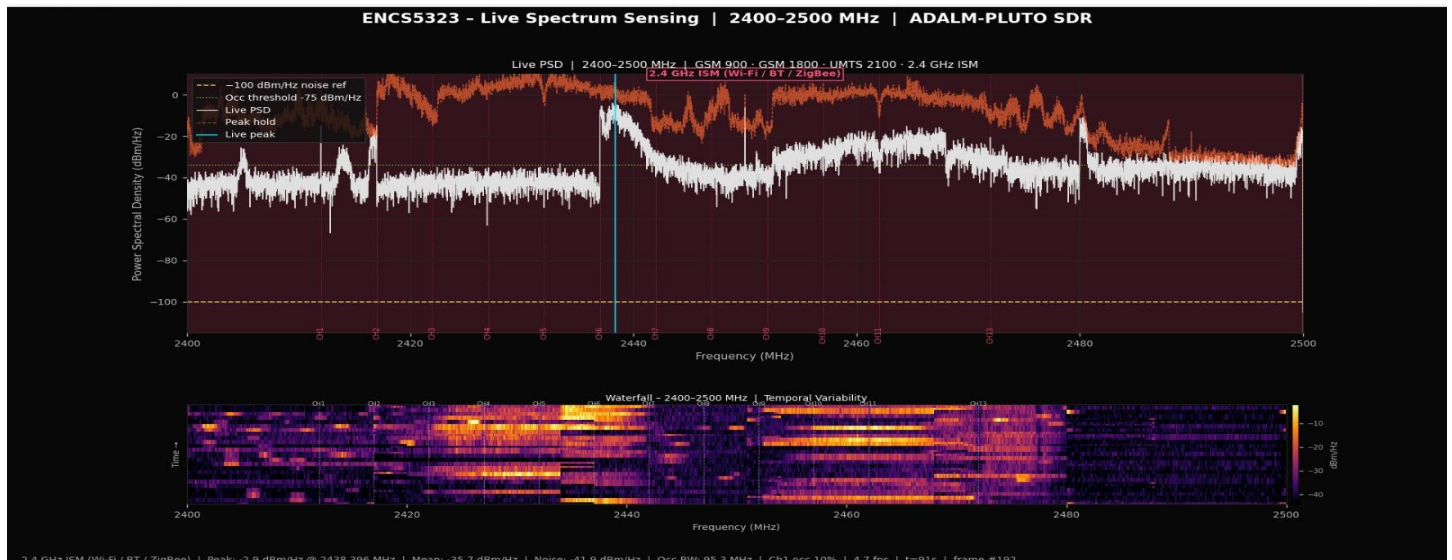


figure 2: Measured WIFI ISM spectrum showing active carrier frequencies and channel occupancy - baseline, no interference

The spectrum changes over time because Wi-Fi sends data in short bursts, so channel activity goes up and down during the measurement.

Figure 2 — Detected Wi-Fi channels (red = occupied)

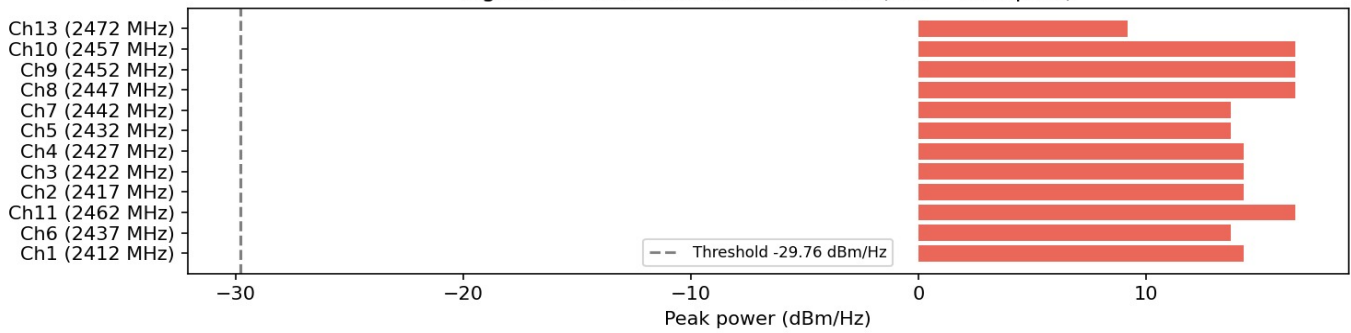


figure 3: Detected Wi-Fi channels (red = occupied).

Figure 3 — Channel occupancy (Ch 1 / 6 / 11)

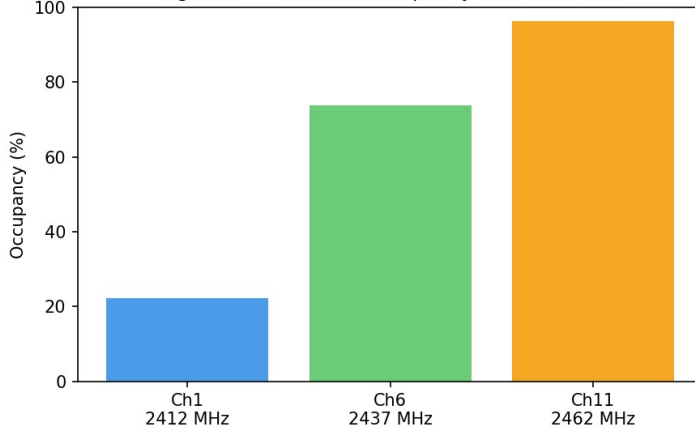


figure 4: Channel occupancy for Ch 1, Ch 6, and Ch 11 (%).

Channel	Centre (MHZ)	Occupancy(%)	Status
1	2412	~22	Light
6	2437	~74	Busy
11	2462	~96	Busy

We measured the 2.4 GHz band (2400–2500 MHz) with the Pluto SDR without the interference generator. Figure 2 shows strong activity between 2410 and 2480 MHz, with peaks near Channel 6 (2437 MHz) and Channel 11 (2462 MHz), and a noise floor of about -38 dBm/Hz. Figures 3 and 4 show normal Wi-Fi channel structure, with Channel 11 busiest (~96%), Channel 6 moderate (~74%), and Channel 1 quietest (~22%).

Component 2 – BW=20Hz, CH1 interference

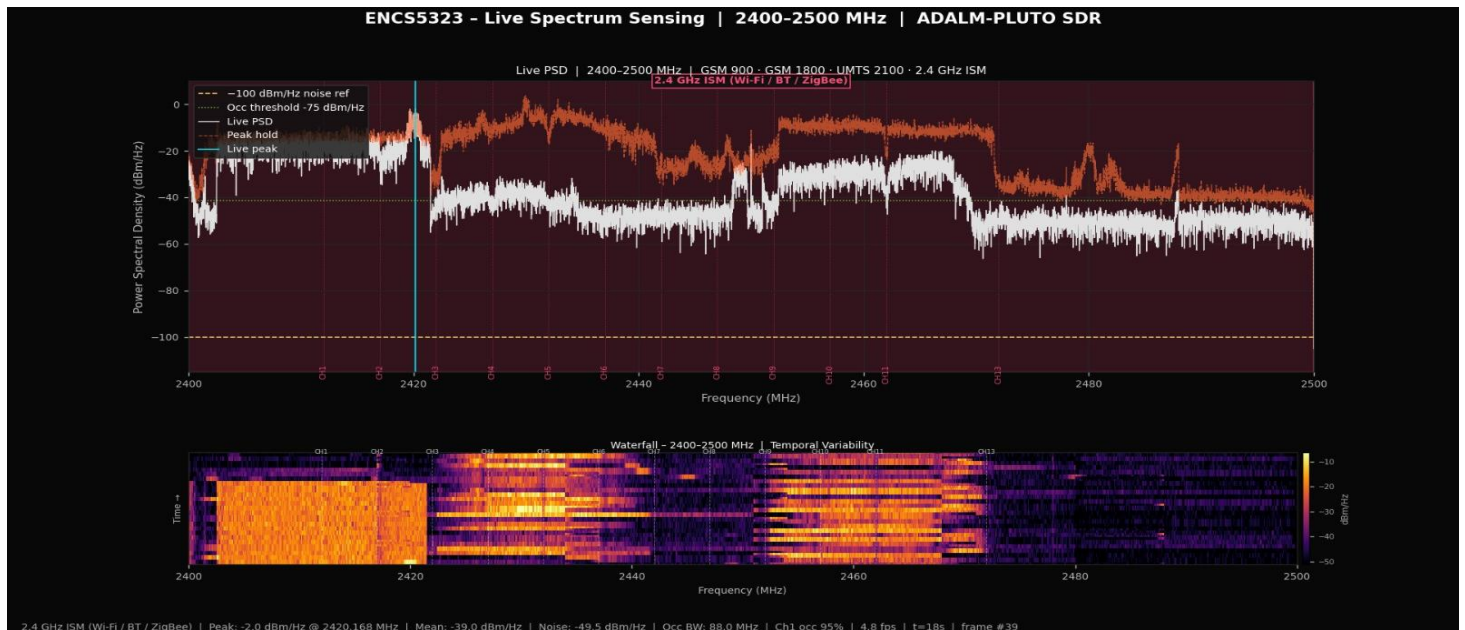


figure 5: Controlled interference at 2412 MHz (Ch 1), BW 20 MHz, gain -10 dB — live spectrum and waterfall.

We used the second Pluto SDR as a transmitter at 2412 MHz (Wi-Fi Channel 1) with 20 MHz bandwidth and -10 dB gain for 45 seconds. The receiver showed a clear power increase in the 2402–2422 MHz range on the waterfall and PSD plots. Channel 1 occupancy rose from about 22% (baseline) to about 97% during this test, with peaks up to 100% while the transmitter was on. Ping to the gateway stayed at 0% loss.

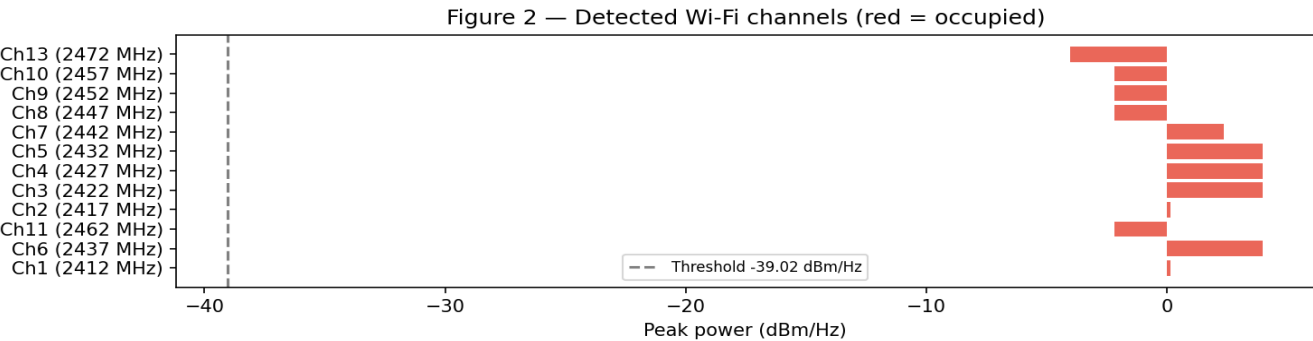
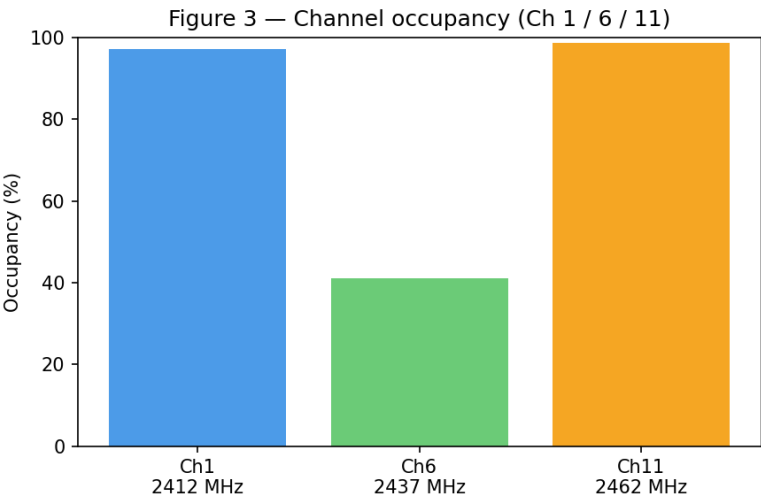


figure 6: Detected channels during interference.(BW=20Hz).



	Baseline	Interference
Ch1 occupancy	~22%	~97%
Bump at 2412 MHz	No	Yes

We transmitted band-limited noise at 2412 MHz (Channel 1) with 20 MHz bandwidth and -10 dB gain for 45 seconds using the second Pluto SDR. The receiver showed a clear power increase at 2402–2422 MHz on the PSD and waterfall (Figure 5-6). Channel 1 occupancy rose to about 97% during the test, compared with ~22% in the baseline (Figure 4). Wi-Fi ping stayed at 0% loss.

figure 7: Channel occupancy during interference (Ch1 ~97%).

Component 2 – BW=10Hz, CH1 interference



figure 8: Controlled interference at 2412 MHz (Ch 1), BW 10 MHz, gain -10 dB — live spectrum and waterfall.

We also tested 10 MHz bandwidth at 2412 MHz (Channel 1) with -10 dB gain for 45 seconds. The PSD shows a narrower bump (~2407-2417 MHz) than the 20 MHz test. Channel 1 occupancy was about 60%, lower than the ~97% seen with 20 MHz. The strongest live peak was still from ambient Wi-Fi on Channel 11.

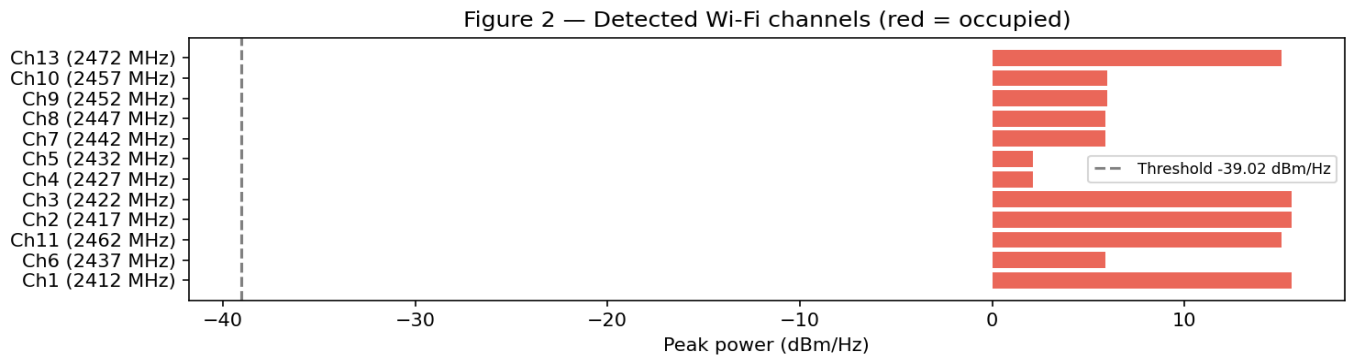
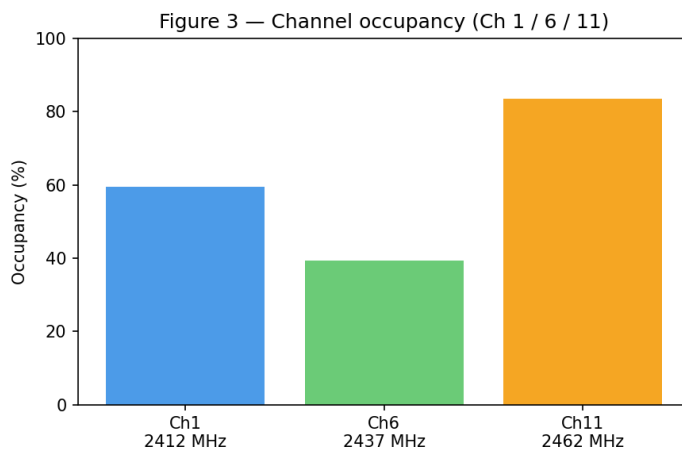


figure 9: Detected channels during interference.(BW=10Hz).



Setting	Bump width	Ch 1 occupancy
BW 20 MHz	~2402-2422 MHz	~97%
BW 10 MHz	~2407-2417 MHz	~60%

Wider bandwidth (20 MHz) produced a stronger and wider interference pattern and higher Channel 1 occupancy than 10 MHz.

figure 10: Channel occupancy — BW 10 MHz (Ch1 ~60%).

Before the transmitter was on, Channel 1 occupancy was about 22%; during interference it increased to about 97% (20 MHz) or about 60% (10 MHz), which shows the spectrum changed when TX was turned on.

Comparing the two interference settings, 20 MHz produced a wider spectral bump (2402-2422 MHz) and higher Channel 1 occupancy (~97%) than 10 MHz (2407-2417 MHz, ~60%), and ping latency was higher under the stronger/wider interference condition.

Interference was generated indoors, at low power (TX gain -10 dB), for 45 seconds per test, aimed only at our group's Wi-Fi equipment in the same room.

Condition	Ping loss	RTT avergae	RTT max
Baseline (no TX)	0%	~3 ms	~7 ms
Interference, BW 20MHz, -10 dB	0%	13.4 ms	188 ms
Interference, BW 10MHz, -10 dB	4%	17.0 ms	557 ms

Compared to baseline, both interference tests increased latency; the 10 MHz run also caused 4% packet loss and a much higher maximum RTT (557 ms vs 188 ms for 20 MHz).